ST FRANCIS CATHOLIC SECONDARY SCHOOL, LAGOS

The Jesuit-Run Catholic College

Basic Science

JSS 1

First Term

2026/2027 SESSION

WEEKS	TOPIC
1	Meaning of science
2	Living and non-living things I
3	Living and non-living things II: Matter
4	Living and non-living things III: Cell
5	Living and non-living things IV: Metals and Non-metals
6	Human Development
7	Mid-term-break
8	Family health
9	Food and nutrition
10-11	Reproductive system I
12	Reproductive system II
13	Examination

Date	
Topic	**MEANING OF SCIENCE**
Objectives/learning outcomes	• Define science and list its major branches; • Explain the scientific methods and its steps; • Appreciate the role of science in solving real world problems; • Identify common laboratory equipment and their uses.
Insight	The word "science" originates from the Latin ***"scientia"***, meaning knowledge, which itself derives from the verb ***"scire"***, "to know," and has roots in Greek ***"epistēmē"***, signifying systematic knowledge. Science is both a body of knowledge and a process. It is exciting, useful and it is ongoing. Science is a global human endeavour.
Key words/New terms	• **Science:** the systematic study of the structure and behaviour of the physical and natural world through observation, experimentation, and the testing of theories against the evidence obtained. • **Scientific methods:** a method of procedure that consists of systematic observation, measurement, and experiment, and the formulation, testing, and modification of hypotheses. • **Hypothesis:** an idea or proposed explanation for a natural phenomenon, formulated based on observations, prior knowledge, or experience, and designed to be testable and falsifiable through experiments or systematic observation • **Theory:** a well-established explanation of natural phenomena based on experimentation and empirical evidence.
Application	• **Medicine (Biology/Chemistry):** Vaccines, disease treatments (e.g., antibiotics). • **Technology (Physics):** Smartphones, GPS, renewable energy (solar panels). • **Agriculture (Earth Science):** Fertilizers, weather forecasting for crop planning. • **Transportation (Physics/Engineering):** Electric cars, aircraft design. • **Environmental Protection (Ecology):** Pollution control, conservation efforts. • **Crime Solving (Forensic Science):** DNA analysis, fingerprint
Summary	• Science is the systematic and logical approach to discovering how things work in the universal. • It is also the study of living/non-living things and natural phenomena to solve problems. • Science has three main branches: **Formal Sciences** (mathematics, logic), **Natural Sciences** (biology, physics, chemistry) and **Social Sciences** (anthropology, economics, psychology, sociology) which are all interconnected. • The scientific method involves observation, hypothesis, experimentation, and theory. • Science improves life (medicine, communication) but can also harm (pollution, weapons).

What I have learnt?	
Skills Acquired	
My Question(s)/ Reflections	i. ii. iii. iv. v.
Action goal	

WEEK ONE

LEARNING ABOUT SCIENCE

Meaning of Science

- Science is defined as the study of living and non- living things in the environment.
- It is also the study of nature to solve human problems.
- Science is the study of the nature and behaviour of natural things and the knowledge that we obtain about them.
- It refers to a system of acquiring knowledge, using observation and experimentation to describe natural phenomena

Fields and Branches of Science

The fields of science are commonly classified along three major lines:

1. **Natural Sciences:** This is the study of the natural world. It is further divided into biological sciences, earth sciences and physical sciences.
2. **Social Sciences:** It is the systematic study of human behaviour.
3. **Formal Sciences:** These include disciplines like mathematics, logic, theoretical computer science, and information theory. They study abstract systems and concepts.

The branches of Natural Sciences can be grouped into:

1. **Biological Sciences** which include Biology and the courses under it: Genetics, Anatomy, Physiology, Zoology, Botany, Microbiology, Biochemistry, Biophysics, Cell Biology, Epidemiology, Ecology, etc.
2. **Earth Sciences** which include Meteorology, Oceanography, Geography, Geology, Environmental science etc.
3. **Physical Sciences** which include Physics and Chemistry and the courses under them: Analytical Chemistry, Organic Chemistry, Inorganic chemistry, Physical Chemistry, Environmental chemistry, Industrial chemistry, Astronomy, Nuclear Physics, Optics, Electromagnetism, Thermodynamics, Quantum Physics, Mechanics, etc.

These branches are interconnected, with knowledge from one area often being essential for progress in others.

Activity 1.0

1. Is Python programming and coding a science subject?
2. Under what branch would you classify it?

Objectives / Aims of Basic science and Technology

1. Develop interest in science and technology.
2. Acquire basic knowledge and skills in science and technology.
3. Become prepared for further studies in science and technology.
4. Avoid drug abuse and related vices.
5. Be safety and security conscious.
6. Take advantage of careers opportunities provided by science and technology.

Scientific methods

A scientific method is defined as the systematic steps scientists follow while carrying out their works.
The steps are as follows:

1. Identifying problems around us e.g., disease outbreak
2. Asking questions
3. Experimentation is scientific or trials.

4. Hypothesis or scientific guess
5. Observation: is close look using sense organs e.g., ear, eye etc.
6. Recording
7. Sorting/Classification
8. Conclusion



Scientific method generally begins this way:

- We observe with our senses on things happening around us;
- We see a pattern which lead to questions or problems which we can try to solve;
- We then put forward an explanation called Hypothesis and then carry out Experiments to test how valid the explanation is to solving the problem;
- We record our experimental observations;
- If our hypothesis is true based on the available evidence, it becomes a Theory. If not, we form a new hypothesis and carry out further experiment until we obtain a correct theory;
- The correct theory is further tested on over time and if remains valid or true it becomes a Law or principle.

Importance of Science

1. For production of food, clothes and shelter.
2. To improve health and medicine for curing diseases.
3. To improve communication.
4. To improve transportation.
5. To discover the problems in our environment such as natural phenomena like floods, earthquakes etc.

Right application of science leads to comfort and better life of people. The right practices of science are:

1. Shelter
2. Production of more food
3. Clothing
4. Purification of water
5. Housing
6. Production of fertilizer
7. Communication
8. Transportation

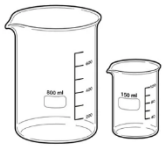
The wrong application of science leads to destruction of life and properties. It includes:

1. Pollution, leading to global warming and ozone layer depletion;
2. Wars, leading to the production of weapons of mass destruction like bombs and missiles, and biological weapons used to destroy lives;
3. Mercy killing;

4. Criminal acts;
5. Drug abuse;
6. Abortion.

Common Equipment & Handling Guidelines

1. **Beaker**



Use: To measure rough estimates (e.g., 250 mL) and mixing liquids.
Handling: Never heat without a Bunsen burner stand.

2. **Graduated Cylinder**



Use: To measure precise liquid volume
Handling: Read at eye level (meniscus); use a funnel to avoid spills when measuring

3. **Pipette**



Use: Transferring small liquid volumes.
Handling: Never mouth-pipette—use a bulb or pump.

4. **Bunsen Burner**



Use: For heating substances.
Handling: Check hose for leaks; Light with striker; adjust air hole for flame type (blue = hottest) and turn off gas immediately after use.

5. **Test Tube**



Use: Use for small scale reactions and heating
Handling: Hold with a test tube holder when heating.**;** point away from people when heating.

6. **Stirring Rod**



Use: for mixing solutions.
Handling: Clean after use to avoid contamination.

7. **Microscope**



Use: To view microscopic specimens.
Handling: Carry with two hands (arm + base); use coarse focus first, then fine focus.

Activity 1.1
List, draw and state the uses of other laboratory equipment not listed

Date	
Topic	**LIVING AND NON-LIVING THINGS I**
Objectives/learning outcomes	• Define living and non-living things. • List at least 5 characteristics of living things (e.g., growth, reproduction, respiration). • Demonstrate curiosity by asking questions about unusual examples (e.g., Is fire living?). • Create a poster classifying examples from their environment. • Define classification and explain its importance in biology • List the seven levels of biological classification • Compare vertebrates and invertebrates with examples
Insight	Understanding the differences between living and non-living things is fundamental in biology. Living things exhibit organized structures, grow, reproduce, respond to stimuli, and undergo metabolism. Non-living things do not exhibit these dynamic processes and lack the internal organization and stability that living things maintain. By recognizing these characteristics, we can appreciate the diversity of life and the intricate balance that sustains ecosystems. Whether it is a towering tree, a playful dog, or a sturdy rock, each element of our environment plays a crucial role in the tapestry of life.
Key words/New terms	• **Taxonomy:** Taxonomy is the science of classification that organizes living organisms into hierarchical categories based on shared characteristics. • **Classification:** This is the activity of assigning objects, plants, animals, etc. into pre-existing classes or categories.
Application	• **Agriculture (Plant Growth & Nutrition):** Crop rotation based on plant life cycles (annual/perennial). • **Medicine (Animal Classification):** Studying vertebrates (mammals) for drug testing. • **Ecology (Adaptation & Competition):** Conservation efforts for endangered species (e.g., tigers). • **Food Industry (Nutrition & Respiration):** Preserving food by inhibiting microbial growth. • **Pest Control (Feeding Habits):** Targeting herbivorous insects (e.g., locusts) to protect crops.
Summary	• Living things (plants, animals, microbes) exhibit traits such as movement, respiration, nutrition, etc. • Plants and animals differ in growth, movement, and food production. • Classification systems (Linnaeus's taxonomy) group organisms by seven levels of classification • Plants are **flowering/non-flowering**; animals are **vertebrates/invertebrates**. They are further classified by feeding habits and life cycles.

What I have learnt?	
Skills Acquired	
My Question(s)/ Reflections	i. ii. iii. iv. v.
Action goal	

WEEK TWO

LIVING AND NON-LIVING THINGS I

Living Things

Living things are things that have life in them. Living things include plants, animals, and microscopic organisms such as bacteria and fungi.

Characteristics of living things (plants and animals)

The characteristics of living things are summed up in the acronym **MR NIGER DAC**.

1. **M for Movement:** Movement is a change in position. Animals move with their total body i.e. from place to place like flying, walking while plants show partial movement like bending.
2. **R for Respiration:** This is where living cells use oxygen to burn food in order to release energy.
3. **N for Nutrition:** This is the taking in of nutrients to get energy.
4. **I for Irritability** (sensitivity): This is the ability of living things to react to changes around them (stimulus) like touch, light, heat, and cold, sound, etc.
5. **G for Growth:** This is the increase in size and height of organisms due to cell multiplication.
6. **E for Excretion:** This is the process of removing waste products of metabolism from the body.
7. **R for Reproduction:** This is the ability to produce or give rise to new individuals of their own kind.
8. **D for Death:** Living things die.
9. **A for Adaptation:** Living things possess certain body features and behaviors to help them survive in their environment.
10. **C for Competition:** Living things struggle for all the necessities of life in order to survive their various environments. They compete for food, water, light, space and mates in order to survive.

Differences between Plants and Animals

Animals	Plants
1. They have limited growth.	They have unlimited growth.
2. They cannot manufacture their own food because they have no chlorophyll.	They can manufacture their own food because they have chlorophyll.
3. They have various body colours.	Most plants are green in colour due to the presence of chlorophyll.
4. They respond to stimuli quickly.	They respond to stimuli slowly.
5. They move from place to place.	They move with their parts but in one spot.

6. They have flexible body structure	They have rigid body structure.
7. Animal cells have various shapes.	Plant cells have definite shape.
8. Food is stored mainly as glycogen and fat.	Food is stored mainly as starch.

Old and Modern System of classification of living things

The science of **Taxonomy**, or the classification of living things, is as old as humanity. People have always found it necessary to be able to organize and identify the living things that are a part of their daily lives. Living things were initially classified as **Plants** and **Animals.**

The first person to formalize the classification of living things was Carl von Linnae, also known as **Linnaeus** (the Latin form), who lived during the 1700's (about 100 years before Darwin). Although his methods were simple and have been expanded upon, some elements of that system are still in use today. He began by dividing all the organisms he knew of into separate **Kingdoms** based on the greatest physical similarities and differences. Subdivisions within Kingdoms were called **Phyla**. Phyla are subdivided into **Classes**. Classes are subdivided into **Orders**. Orders are subdivided into **Families**. Families are subdivided into **Genera**. Finally, the Genera are subdivided into **species**.

Here are a few examples:

Common Name	**Human**	**Chimpanzee**	**Lynx**	**Blue Jay**	**Honeybee**	**Octopus**	**Sunflower**
Kingdom	Animalia	Animalia	Animalia	Animalia	Animalia	Animalia	Plantae
Phylum	Chordata	Chordata	Chordata	Chordata	Arthropoda	Mollusca	Tracheophyta
Class	Mammalia	Mammalia	Mammalia	Aves	Insecta	Cephalopoda	Angiospermae
Order	Primata	Primata	Carnivora	Passeriformes	Hymenoptera	Octopoda	Asterales
Family	Hominidae	Hominidae	Felidae	Corvidae	Apidae	Octopodidae	Asteraceae
Genus	*Homo*	*Pan*	*Lynx*	*Cyanocitta*	*Apis*	*Octopus*	*Helianthus*
species	*sapiens*	*troglodytes*	*lynx*	*cristata*	*mellifera*	*vulgaris*	*annuus*

Classification of plants and animals using different criteria (life cycles, uses, feeding habits)

Classification of Plants

Plants are grouped into two:

1. Flowering plants;

2. Non-flowering plants.

1. **Flowering plants:** These are plants that bear flowers e.g. okra plant, maize plant, orange tree, iroko tree, hibiscus flower, rose flower, etc.



2. **Non-flowering plants:** They are plants that do not bear flowers e.g. Mushroom, yeast, bread mould (Rhizopus), spirogyra (algae), mosses, fern, etc.



Other classifications of plants include:

1. **Classification based on their life cycle** e.g. annual plants (plants that complete their life cycle in one year), biennial plants (plants that complete their life cycle in two years), and perennial plants (plants that complete their life cycle in more than two years).
2. **Classification based on their growth habits** e.g. herbs, shrubs, trees, climbers and creepers.
3. **Classification based on their uses** e.g. legumes, cereals, fruits, vegetables, spices, roots and tubers, etc.

Classification of animals

Animals are grouped into two:

1. Vertebrate animals;
2. Invertebrate animals.

Vertebrates are animals that have backbones.

Classes of Vertebrates include:

1. **Pisces** e.g. all kinds of fish like sharks, rays, catfish, tilapia, mackerel, herring, sardine, etc.
2. **Amphibians** e.g. newt, salamanders, frogs, toads, etc.
3. **Reptiles** e.g. lizards, chameleons, snakes, crocodiles, wall geckos, etc.
4. **Aves** e.g. all kinds of birds like ostriches, kiwis, chickens, eagles, kites, kingfishers, etc.
5. **Mammals** e.g. dogs, cats, duck-billed platypuses, whales/orcas, monkeys, man, etc.

Invertebrates are animals that do not have backbones.

Classes of Invertebrates include:

1. **Protozoa** e.g. plasmodium, amoeba
2. **Porifera** e.g. sponges
3. **Coelenterata** e.g. sea anemones, hydra
4. **Platyhelminthes** e.g. tapeworms, liver flukes
5. **Nematoda** e.g. nematodes like hookworms, pinworms and whipworms
6. **Annelida** e.g. earthworms, leeches
7. **Echinodermata** e.g. starfish, sea urchins
8. **Mollusca** e.g. octopus, snails, slugs
9. **Arthropoda**: This is the largest class of invertebrates. It is subdivided into:

a. **Insects** (butterflies, cockroaches, houseflies, mosquitoes)
b. **Arachnids** (spiders, mites and scorpions)
c. **Crusteaceans** (prawns, lobsters, crabs)
d. **Myriapods** (centipedes and millipedes)

Other classifications of animals include:

1. **Classification based on their feeding habits:** e.g. herbivores (animals that eat only plants), carnivores (animals that eat only meat) and omnivores (animals that eat both meat and plants).
2. **Classification based on their life cycle** e.g. oviparous animals (animals that lay eggs) and viviparous animals (animals that give birth to their young alive).

Activity 2.0

From the lists of vertebrates and invertebrates listed above and their properties; in a tabular form, state five differences between a vertebrate and invertebrate.

Date	
Topic	**LIVING AND NON-LIVING THINGS II: MATTER**
Objectives/learning outcomes	• Define matter, mass, weight; • Differentiate between mass and weight; • Define the terms involve in the states of matter; • Classify examples of matter into their respective states.
Insight	The concept of matter is fundamental to understanding the physical world. Everything around us-air, water, our body, books, etc. is made up of matter. It is the building block of all existence, from the smallest subatomic particles to the largest galactic structures.
Key words/New terms	• **Matter:** Any substance that has mass and takes up space. • **Mass:** Mass is the amount of matter in an object. • **Weight:** Weight is the force of gravity acting on a body. • **Density:** The mass of a unit volume of a material substance. • **Volume:** Volume is the three-dimensional space occupied by a substance or enclosed by a surface.
Application	• **Cooking (Melting/Freezing)**: Ice melting into water, butter solidifying when chilled. • **Weather (Condensation/Evaporation)**: Cloud formation (gas → liquid), drying clothes (liquid → gas). • **Industry (Sublimation/Deposition)**: Dry ice for cooling (solid → gas), mirror coating (gas → solid). • **Energy (Plasma/Ionization)**: Neon lights (ionized gas), solar panels (sun's plasma energy). • **Transport (Gas Compression)**: LPG fuel (compressed gas → liquid for storage). • **Medicine (Sterilization)**: Plasma used to disinfect surgical tools.
Summary.	• Matter is anything with mass and volume, existing in solid, liquid, gas, and plasma states. • Mass (constant, measured in kilogram) differs from weight (varies, measured in Newton). • States of matter have unique properties (shape, density, energy). Changes between states (melting, vaporization, ionization) occur due to energy exchange.

What I have learnt?	
Skills Acquired	
My Question(s)/ Reflections	i. ii. iii. iv. v.
Action goal	

WEEK THREE
LIVING THINGS II: MATTER

Matter is anything that has mass and occupies space. Almost everything in the universe is made up of matter.
Mass differs entirely from weight in the following:

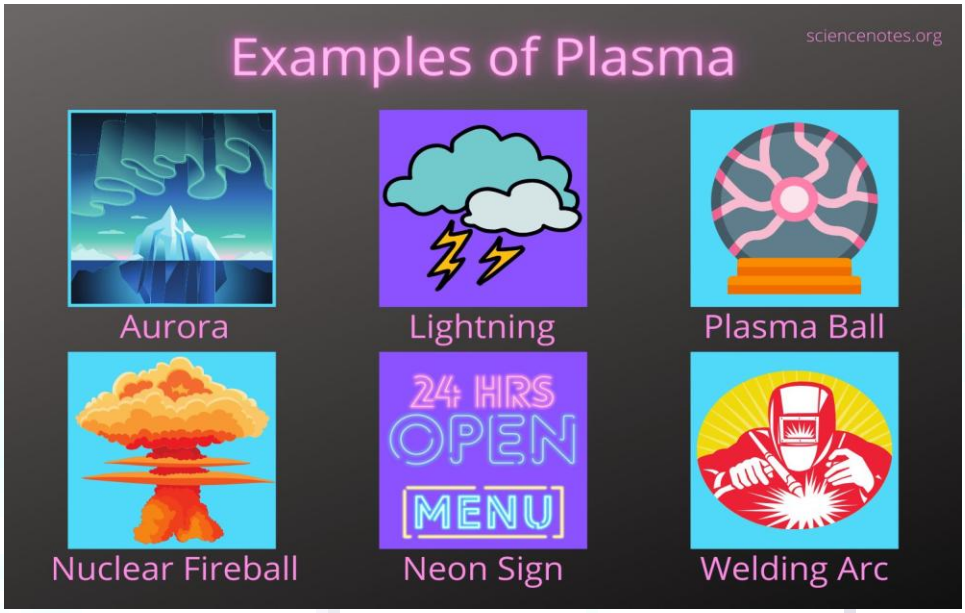
Mass	**Weight**
Mass is the quantity of matter in an object	Weight is the force that attracts an object to the center of the earth.
The mass of a body is usually constant.	The weight of a body varies across the surface of the earth.
Mass is measured in kilograms (kg).	Weight is measured in Newton (N).
The instrument used to measure the mass of a body is the beam or chemical balance.	The instrument used to measure the weight of a body is the spring balance.

Properties of matter include color, odor, mass, volume, temperature, and texture.
There are four fundamental states of matter: solid, liquid, gas and plasma. These are the ones that occur naturally in the universe.

PROPERTIES OF THE STATES OF MATTER

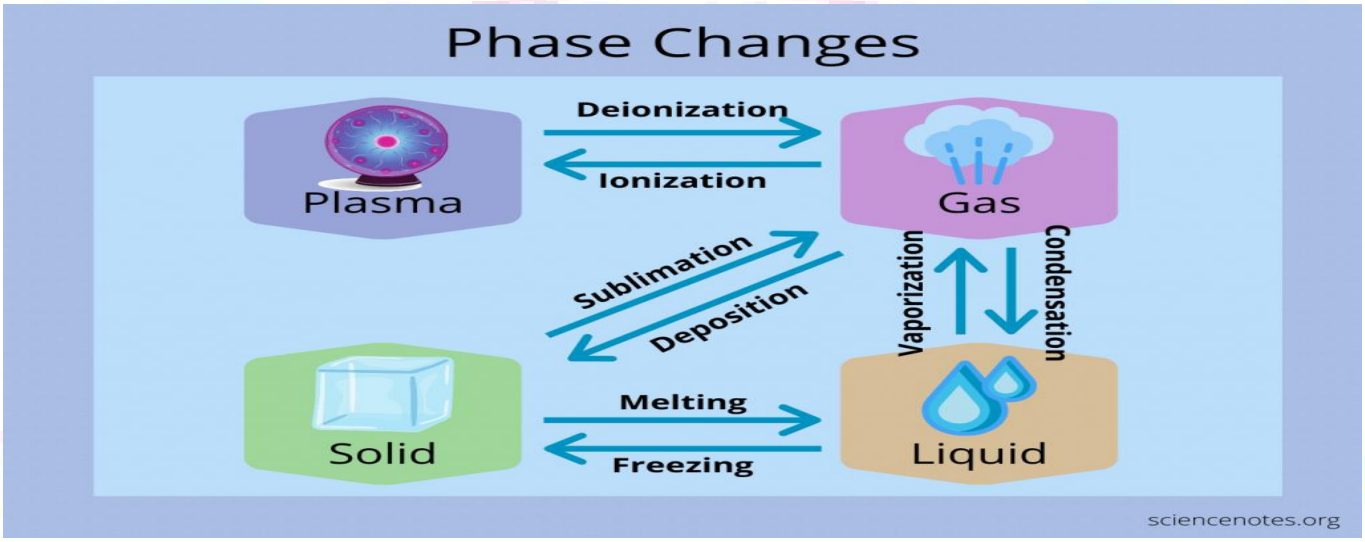
Property	**Solid**	**Liquid**	**Gas**
Volume and shape	They have definite volume and definite shape.	They have definite volume, but not definite shape. They take the shape of their containers.	They have neither definite volume nor definite shape.
Rigidity and Compressibility	They are very rigid and not compressible	They are less rigid and no compressible	They are the least rigid and highly compressible
Density	They have high density	They have less density than solids	They have the least density
Intermolecular force of attractions	There is a strong force of attraction between the particles of solids	The force of attraction in liquids is weaker than those of solids	There is little to no force of attractions in gases
Kinetic Energy	Least kinetic energy	More kinetic energy than solids	They have the most kinetic energy
Examples	Stone, ice, wood, iron, copper, diamond, gold, etc.	Mercury, water, alcohol, petrol, kerosene, etc.	Nitrogen, oxygen, carbon(iv)oxide, helium, hydrogen, etc.

PLASMA



Like gases, plasmas have no fixed shape or volume, and are less dense than solids or liquids. But unlike ordinary gases, plasmas are charged gases, with strong electrostatic interactions.
1. Because it consists of charged particles, plasma is a good conductor of electricity, unlike ordinary gases.
2. Because the charged particles have kinetic energy, plasma always has a magnetic field.
3. Plasma is more readily influenced by electric and magnetic fields than by gravity.
4. Because of the totally chaotic and highly energetic state of the constituent particles of plasma, it produces its own electromagnetic radiation.
Examples of plasma include the sun, stars, auroras, comet tails, lightning, solar winds, neon signs, etc.

CHANGES IN THE STATE OF MATTER



1. **Melting**: This is the change of the state of matter from solid to liquid. For example, solid ice melts into liquid water.

2. **Freezing**: This is the change of the state of matter from liquid to solid. Freezing changes water from a liquid into solid ice.

3. **Vaporization:** This is the change of the state of matter from liquid to gas. An example is the evaporation of rubbing alcohol or petrol from skin into the air.

4. **Condensation:** This is the change of the state of matter from gas to liquid. A good example of condensation is dew formation (or cloud formation) from water vapor in air.

5. **Sublimation:** This is the change of the state of matter from solid directly to gas. Dry ice undergoes sublimation to change from solid carbon dioxide directly into carbon dioxide gas. Another example is the transition from ice directly into water vapor on a cold, windy winter day.

6. **Deposition**: This is the change of the state of matter from gas to solid. Hoarfrost is grayish-white frost that forms during clear, cold weather when water vapor deposits as ice. Another example is deposition of silver vapor onto glass to form a silver mirror.

7. **Ionization**: This is the change of the state of matter from gas to plasma. When you turn on a plasma ball toy, the noble gases inside are ionized by an electric charge and become plasma. The aurora is another example of ionization.

8. **Deionization or Recombination:** This is the change of the state of matter from plasma to gas. Lightning is an example of plasma. After a lightning strike, nitrogen ions eventually draw closer together and lose their charge to become nitrogen (N_2) gas.

Date	
Topic	**LIVING AND NON-LIVING THINGS III: CELLS**
Objectives/learning outcomes	• Define a cell as the basic unit of life. • Identify and label the main parts of plant and animal cells (e.g., nucleus, cytoplasm, cell membrane, chloroplast, cell wall). • Compare and contrast plant and animal cells • Draw and label a plant cell and an animal cell with correct structures
Insight	Cells are the fundamental building blocks of life, serving as the basic units of structure and function in all living organisms. All living things-plants, animals, Fungi and even bacteria-are made of cells. The study of cells has led to significant advancements in medical research, biotechnology, and healthcare, with applications ranging from cancer therapy to regenerative medicine.
Key words/New terms	• **Cell:** A cell is the basic structural and functional unit of all living organisms, capable of carrying out essential life processes. • **Nucleus:** The cell nucleus is a membrane-bound organelle that stores genetic material and regulates cellular activities in eukaryotic cells. • **DNA (Deoxyribonucleic acid):** The molecule that carries the genetic instructions for the development, functioning, and reproduction of all living organisms. • **Ribosomes:** Ribosomes are cellular machines responsible for synthesizing proteins. • **Mitochondria:** Mitochondria are organelles often referred to as **"the powerhouses"** of the cell because they generate most of the cell's energy. • **Cell membrane:** The cell membrane is a selectively permeable barrier that surrounds all living cells, regulating transport, communication, and structural support.
Application	• **Medicine (Nucleus/DNA)**: Genetic testing for diseases. • **Agriculture (Chloroplasts)**: Crop breeding for better photosynthesis. • **Energy (Mitochondria)**: Studying cellular respiration to improve human stamina. • **Food Science (Vacuoles)**: Starch storage in potatoes (plant cells) vs. glycogen in meat (animal cells). • **Biotechnology (Cell Membrane)**: Drug delivery systems using semi-permeable membranes. • **Ecology (Cellulose Cell Wall)**: Biofuel production from plant cell walls.
Summary.	• Cells are the basic unit of life, containing organelles that perform vital functions. • Plant and animal cells share components (nucleus, mitochondria) but differ in structure (cell wall, chloroplasts). • The nucleus controls cell activities, mitochondria produce energy, and chloroplasts enable photosynthesis. • Cells determine an organism's functions and characteristics.

What I have learnt?	
Skills Acquired	
My Question(s)/ Reflections	i. ii. iii. iv. v.
Action goal	

WEEK FOUR

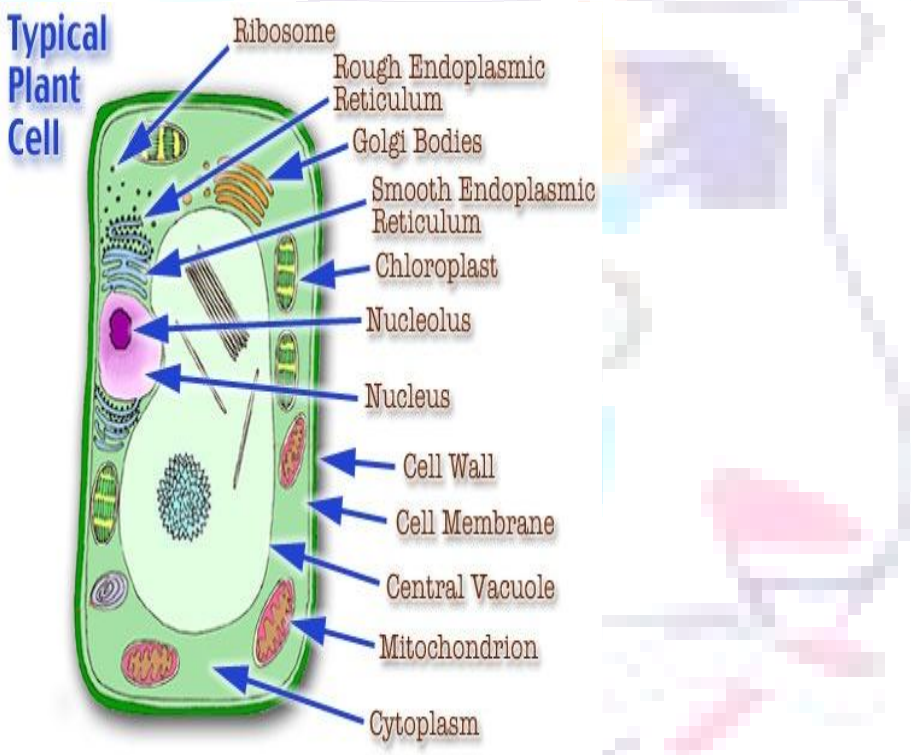
LIVING THINGS III: CELLS

Living things have cells. These cells make them to be similar and different in many ways.

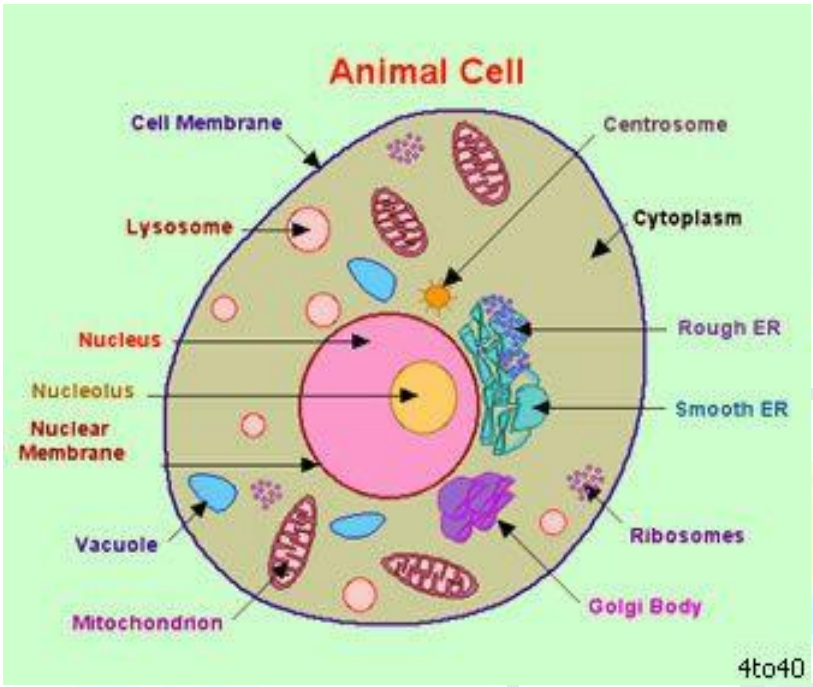
Most cells are so small that you can only see them with a microscope. Cells have different components and each component performs its own function within the cell.

A cell is defined as **the smallest unit of life** or **the basic unit of life**.

Plant Cell



Animal Cell



Similarities between plant cell and animal cell

1. Presence of cytoplasm
2. Presence of nucleus
3. Presence of mitochondria
4. Presence of golgi bodies
5. Presence of lysosome.

Differences between plant and animal cell

Animal cell	Plant cell
No cellulose cell wall	Has cellulose cell wall
Has no definite shape	Has a definite shape or rigid
Has no chloroplasts	Has chloroplasts
Has smaller vacuole	Has larger vacuole
Stores carbohydrate as glycogen	Stores carbohydrate as starch

FUNCTIONS OF THE CELL PARTS/ ORGANELLES

1. **Nucleus:** It is the spherical structure found mainly at the center of a cell. It controls most of the cellular and chemical activities of the cell. It contains DNA, the genetic information that cells need to grow and reproduce.

2. **Cytoplasm:** It is the largest part of the cell. It contains all the structures of the cell like the nucleus, vacuole etc.

3. **Food vacuole:** it is where the digestion and absorption of food takes place. In plants, they manufacture their own food and store their food (carbohydrate) as starch while animals do not manufacture food but depends on plant for their food. They store their carbohydrate as glycogen.

4. **Cell membrane**: It is a semi- permeable membrane that covers the cell. It allows water, salt, proteins, and glucose to pass in and out of the cell.

5. **Cellulose cell wall**: it is the outermost part of a plant cell. The cell wall is made of tough material called cellulose. It gives rigidity to plant cell.

6. **Chloroplasts**: They are found only in plant cells. They are the site for photosynthesis since they contain the green pigment called the chlorophyll.

7. **Mitochondria**: This is the power house of the cell. It is the site for respiration where oxygen burns up food to produce energy for the cell.

Date	
Topic	**LIVING AND NON-LIVING THINGS IV: METALS AND NON-METALS**
Objectives/learning outcomes	• Compare physical properties between metals and non-metals • Explain why certain metals are used for specific purpose • Conduct simple test to distinguish a metal from a non-metal • Give examples of common metals and non-metals
Insight	Non-living matter are classified into three groups- metals, semi-metals and non- metal. Their uses depend on their nature/properties.
Key words/New terms	• **Metals:** Metals are elements or alloys that conduct heat and electricity efficiently and typically have a lustrous, shiny surface when polished or fractured. • **Non-metals:** Nonmetals are chemical elements that lack metallic properties, are poor conductors of heat and electricity, and tend to gain electrons in chemical reactions. • **Semi-metals:** Semimetals, also known as metalloids, are elements with properties intermediate between metals and nonmetals, often acting as semiconductors.
Application	• **Construction:** Metals (iron, steel) for bridges, buildings. • **Electronics**: Copper/Aluminum (wires) conduct electricity. • **Healthcare:** Oxygen (respiration), Sulfur (antibiotics). • **Agriculture:** Nitrogen/Phosphorus (fertilizers). • **Energy**: Uranium (nuclear power), Carbon (batteries). • **Water Treatment**: Chlorine (disinfection).
Summary.	• Metals (e.g., iron, gold) are hard, shiny, conductive, and form positive ions, while non-metals (e.g., oxygen, sulfur) are dull, brittle, insulating, and form negative ions. • Metals are used in construction, electronics, and jewelry, while non-metals are vital for medicine, fertilizers, and respiration. • The chemical reactions of metals differ from those of non-metals (metals displace hydrogen; non-metals form acids).

What I have learnt?	
Skills Acquired	
My Question(s)/ Reflections	i. ii. iii. iv. v.
Action goal	

WEEK FIVE

LIVING AND NON-LIVING THINGS IV: METALS AND NON-METALS

Non-living things are things that have no life in them. They are usually in the form of **solids, liquids**, **gases** and **plasma**.

Non-living things are grouped into two:

1. **Metals:** They lose electrons to become positively charged.
2. **Non-metals:** They gain electrons to become negatively charged.

The properties of metals and non-metals are used to differentiate them.

Physical Properties of metals and non-metals

Metals	Non metals
They are hard	They are soft
They have high densities	They have lower densities
They are malleable	They are not malleable
They are ductile	They are not ductile
They have high melting and boiling points	They have low melting and boiling points
They are lustrous i.e. shiny and polish able	They are neither lustrous nor shiny, but dull
They are good conductors of heat and electricity	They are bad conductors of heat and electricity.

Chemical properties of metals and non-metals

Metals	Non metals
They produce positively charged ions	They produce negatively charged ions
They are reducing agents by losing electrons	They are oxidizing agents by gaining electrons.
They produce basic oxides	They produce acidic oxides
They displace hydrogen from acids to form salts	They do not displace hydrogen in acids to form salts.

Uses of metals

1. They are used for making jewels.
2. They are used for manufacturing cooking utensils, doors, aircrafts, etc.
3. They are used for constructing auto mobiles, buildings, bridges etc.
4. They are used for generating nuclear energy.
5. They are used for producing electrical wires and plumbing materials.

Uses of non-metals

1. They are used in medicine production e.g. sulphur
2. They are used for producing fertilizers e.g. nitrogen
3. They are used for making car batteries e.g. carbon
4. They are used for making laboratory chemicals e.g. sulphuric acid, lime water, etc.
5. They are used in respiration like oxygen.
6. They are used in disinfecting water and for making bleaching powder.

Examples of metals:

Gold, silver, aluminum, iron, potassium, sodium, copper, lead, mercury, zinc, tin, calcium etc

Examples of non-metals:

Oxygen, carbon, nitrogen, sulphur, chlorine, phosphorus, iodine, fluorine

Date	
Topic	**HUMAN DEVELOPMENT**
Objectives/learning outcomes	• Define puberty and adolescence and state their age ranges. • Identify physical, emotional, and social changes during puberty. • Explain the menstrual cycle and its importance in reproduction. • List personal hygiene practices essential during puberty
Insight	Human development is the study of how people grow and change, physically, mentally, and socially, from conception through the end of life. It's not just about childhood. The field tracks every shift in the body, thinking, and relationships across every stage the person will pass through, from the rapid changes of infancy to the slower but significant transformations of late adulthood. Studying human development will give one a greater appreciation of his/her own development throughout life.
Key words/New terms	• **Puberty:** The period during which growing boys or girls undergo the process of sexual maturation. • **Adolescence:** A transitional phase of human development that occurs between childhood and adulthood, typically defined as ages 10 to 19. It is the stage when children develop into adults, and these physical changes are caused by hormonal changes in the body.
Application	• **Health Education:** Teaching menstrual hygiene to prevent infections (e.g., proper pad disposal). • **Parenting:** Understanding mood swings in teens for better communication. • **Medicine:** Hormone therapy for delayed puberty. • **Psychology:** Counseling adolescents on body image and emotional changes. • **Fitness:** Tailoring exercise for growing bodies (e.g., avoiding heavy weights in early puberty).
Summary	• Puberty is the transition from childhood to adolescence (girls: ~11 yrs, boys: ~12 yrs), marked by secondary sexual characteristics (voice changes, menstruation, hair growth). • Adolescence involves physical, social, and emotional changes (mood swings, independence). • Hormones (testosterone in males, estrogen/progesterone in females) drive development. • Personal hygiene (sanitary pads, clean clothing) is crucial to prevent infections.

What I have learnt?	
Skills Acquired	
My Question(s)/ Reflections	i. ii. iii. iv. v.
Action goal	

WEEK SIX

HUMAN DEVELOPMENT

Puberty and Adolescence

Puberty is the period or stage at which male and female sex organs mature. It is also a period of transition from childhood to adolescence. It is characterized by the development of secondary sexual characteristics. It starts from the age of 11 for girls and 12 for boys.

Adolescence is the period of development between childhood and adulthood. Certain hormones are responsible for the development of male and female during puberty. These hormones are testosterone in male and estrogen and progesterone in female.

Physical changes during puberty/adolescence

Physical Changes in Boys During Adolescence	Physical Changes in Girls During Adolescence
Increase in height and body size	Increase in height and body size
Shoulders become broader	Hips become wider
Muscles develop and become stronger	Breasts begin to develop
Growth of hair on the face (beard and moustache), chest, armpits, and pubic region	Growth of hair in the armpits and pubic region
Voice deepens and becomes heavier	Voice becomes slightly higher and more mature
Penis and testes enlarge	Ovaries and uterus mature
Increased sweating and oily skin	Increased sweating and oily skin
Production of sperm begins and "wet dreams"	Menstruation (monthly periods) begins

Social and Emotional Changes during Adolescence

Both boys and girls experience several social and emotional changes during adolescence, including:

1. Increased awareness of their appearance and self-image;
2. Mood swings due to hormonal changes;
3. A desire for greater independence from parents and guardians;
4. Stronger interest in friendships and spending time with peers;
5. Development of feelings of attraction toward others;
6. Increased sensitivity to others' opinions and emotions;
7. Greater ability to think about the future and make personal decisions;
8. Development of a stronger sense of identity and self-confidence.

These changes help adolescents transition from childhood to adulthood and learn how to interact with others and manage their emotions.

Personal hygiene during Puberty/Adolescence

Personal hygiene is the act of keeping the body neat and healthy. Personal hygiene during puberty must be maintained to avoid body odor and diseases. Important habits include:

1. Bathing regularly to keep the body clean and remove sweat and dirt.

2. Washing the face daily to reduce excess oil and help prevent pimples.
3. Brushing teeth at least twice a day and maintaining good oral hygiene.
4. Washing hair regularly to keep it clean and healthy.
5. Using deodorant or antiperspirant if needed to control body odor.
6. Wearing clean clothes and changing underwear daily.
7. Keeping nails clean and trimmed to prevent the spread of germs.
8. Washing hands frequently, especially before eating and after using the toilet.
9. **Girls:** Maintaining proper menstrual hygiene by changing sanitary products regularly and keeping the genital area clean.
10. **Boys:** Keeping the genital area clean and practicing good personal care.

Terms used in Puberty and Adolescence

1. **Menstruation:** This is the monthly discharge of blood from the vagina. It lasts 3-5 days and is called menses
2. **Menstrual cycle:** This is the beginning of one period to the beginning of the next cycle i.e. 28 days
3. **Ovulation:** This is the release of eggs from the ovary.
4. **Wet dreams:** This is the release of seminal fluid from a boy's penis during sleep.

Stages in human development are:

1. Infancy: (0-2years)
2. Childhood: (2-12years)
3. Adolescence: (12-18years)
4. Adult hood: (18-above)

WEEK SEVEN: MID-TERM BREAK

Date	
Topic	**FAMILY HEALTH (PERSONAL HYGIENE/CLEANLINESS)**
Objectives/learning outcomes	• Define personal hygiene and environmental cleanliness • List essential personal hygiene practices • Identify methods of maintaining environmental cleanliness • Explain health consequences of poor hygiene and sanitation
Insight	Good hygiene promotes health, confidence, and well-being.
Key words/New terms	**Sanitation:** The conditions that promote health by preventing human contact with the hazards that come from waste. **Hygiene:** conditions or practices conducive to maintaining health and preventing disease, especially through cleanliness.

Application	• **Disease Prevention**: Handwashing reduces COVID-19, cholera, and diarrhea spread. • **Mental Health**: Clean homes improve mood and productivity. • **Social Acceptance**: Fresh breath (brushing) and clean clothes boost confidence in public. • **Child Development**: Schools teach hygiene to prevent lice, scabies, and infections. • **Public Health**: Proper sewage disposal prevents malaria and typhoid outbreaks.
Summary	• Family health involves personal hygiene (body, clothes, teeth cleaning) and home sanitation (clean floors, toilets, furniture) to prevent diseases. • Poor hygiene leads to illness, bad odor, and social isolation. • Regular washing, nail trimming, and flushing toilets maintain cleanliness. • Good hygiene promotes health, confidence, and well-being.
What I have learnt?	
Skills Acquired	
My Question(s)/ Reflections	i. ii. iii. iv. v.
Action goal	

WEEK EIGHT

FAMILY HEALTH (PERSONAL HYGIENE/CLEANLINESS)

Sanitation means cleanliness; it is the hygienic means of promoting good health through prevention of human contact with the hazards of wastes as well as the treatment and proper disposal of sewage wastewater.
Personal hygiene can be described as the practice of keeping one's body, clothes, and shoes in a clean and neat condition all the time.

Ways of keeping our body clean

1. Washing our body with water, soap and sponge at least twice a day; morning and night as well as after some games and sporting activities
2. Boys should cut their hair short an girls should wash their hair regularly
3. Use a nail clipper to always cut the nails short
4. Wash your teeth twice day
5. The hands should be kept clean by washing them with clean running water
6. Wash your clothes immediately after wearing them

Ways of keeping our home and environment clean

1. Sweep and mop floors regularly to remove dust and dirt.
2. Dispose of waste properly in bins and dustbins.
3. Do not litter the surroundings with rubbish.
4. Wash dishes and utensils after use.
5. Clean toilets and bathrooms regularly.
6. Keep drains and gutters free from blockage.
7. Trim grasses and clear bushes around the home.
8. Cover waste bins to prevent flies and bad odors.
9. Plant trees and flowers to beautify and protect the environment.
10. Participate in community clean-up activities.
11. Recycle and reuse materials where possible.
12. Avoid dumping refuse into rivers, streams, or drainage channels.
13. Ensure proper ventilation by opening windows and keeping rooms tidy.
14. Clean furniture, walls, and windows regularly.
15. Educate family members and neighbors on the importance of cleanliness.

Consequences of poor hygiene

1. Spread of diseases such as cholera, typhoid, diarrhea, and skin infections.
2. Bad body odor due to the accumulation of sweat and bacteria.
3. Tooth decay and gum disease from poor oral hygiene.
4. Skin problems such as rashes, acne, and fungal infections.
5. Infestation by pests like flies, mosquitoes, rats, and cockroaches.
6. Contamination of food and water, leading to illness.
7. Unpleasant living environment with bad smells and dirt.
8. Low self-esteem and embarrassment due to poor personal appearance.
9. Poor health and reduced productivity because of frequent sickness.
10. Environmental pollution caused by improper waste disposal.

Date	
Topic	**FOOD AND NUTRITION**
Objectives/learning outcomes	• Define nutrition, nutrients, and balanced diet • Identify classes of food and their main functions • Explain the consequences of malnutrition (over-nutrition and under-nutrition) • Classify given food items into their respective nutrient groups • Prepare a simple healthy snack using different food groups
Insight	The nutrition industry is increasingly focused on functional foods and ingredients that provide targeted health benefits, such as supporting gut health, protein intake, and cognitive performance. Food in itself is medicine. We are what we eat. What we eat directly impacts our bodies and our mind.
Key words/New terms	**Food:** Any substance that people eat or drink to provide energy, support growth, and maintain good health. **Nutrition:** The process by which the body takes in and uses food nutrients for growth, energy, repair, and proper body functioning.
Application	• **Health & Fitness**: Athletes consume high-protein diets for muscle repair and carbs (carbohydrates) for energy. • **Disease Prevention**: Vitamin C (fruits) boosts immunity; fiber (oats, legumes) aids digestion. • **Child Development**: Calcium (dairy) strengthens bones; iron (meat, spinach) prevents anemia. • **Weight Management**: Balancing fats (good vs. bad) and fiber helps control obesity. • **Hydration**: Water regulates body temperature and detoxifies (e.g., post-workout recovery).
Summary.	• Food is any substance consumed to sustain life, provide energy, and support growth, while nutrition is how the body converts food into energy and tissues. • The five food groups (vegetables, protein, fruits, grains, dairy) supply essential nutrients (carbs, proteins, fats, vitamins, minerals, water, fiber). • Each nutrient has specific functions, like energy (carbs), tissue repair (proteins), and disease prevention (vitamins). A balanced diet ensures proper body function and health.

What I have learnt?	
Skills Acquired	
My Question(s)/ Reflections	i. ii. iii. iv. v.
Action goal	

WEEK NINE

FOOD AND NUTRITION

Food is any nourishing substance that is eaten, drunk, or otherwise taken into the body to sustain life, provide energy and promote growth. **Nutrition** is the process by which the body nourishes itself by transforming food into energy and body tissues.

Types of Food

There are 5 major food groups:

- Vegetables
- Protein
- Fruits
- Grains
- Dairy.

Food provides essential chemical substances called **nutrients**. The body needs these nutrients to help it make energy; to grow, repair, and maintain its tissues; and to keep its different systems working smoothly.

Classes of Food, Examples and Functions

Classes of Food	Examples	Functions
Carbohydrates	Rice, yam, bread, maize, guinea corn, millet, etc.	They provide the body with energy.
Proteins	Beans, fish, eggs, milk, meat, etc.	They build, repair and maintain healthy body tissues.
Fats and Oils	Vegetable oil, palm oil, animal fat, butter, margarine	Fats serve as a kind of energy source. They prevent heat loss in extreme cold weather and protect organs against shock. They are responsible for making up part of our body cells and transporting fat-soluble vitamins such as vitamin A, D, E and K.
Vitamins	Fruits and vegetables, yoghurt, cheese	They protect the body against diseases, maintain healthy skin

		and hair, build bones and release and utilize energy from foods.
Minerals	Fruits and vegetables, milk and dairy products, fish, unripe plantain, nuts, mushrooms, etc.	They regulate many body functions such as fluid balance, muscle contraction and transmission of nerve impulses. Some minerals are the structural component of the body, such as calcium in bones and teeth.
Water	Water, any food or drink containing water	It regulates body temperature, produces body fluids, transports nutrients and removes waste products
Dietary fibre	Legumes, oats, oranges, apples, bananas, green peas, etc.	It helps stabilize blood sugar, promote gastrointestinal health and prevent constipation.

Date	
Topic	**REPRODUCTIVE SYSTEM**
Objectives/learning outcomes	• Differentiate between asexual and sexual reproduction in plants • Identify structures involve in sexual and asexual reproduction • State 3 functions each for male and female reproductive organs • Describe processes like pollination, fertilization and vegetative propagation Appreciate the role of pollinators in the ecosystem
Insight	Reproduction is a complex process involving various biological and genetic factors. Recent research has provided valuable insights into reproductive biology, genetics, and medicine. The reproductive system is the biological system responsible for producing offspring, it includes both internal and external organs.
Key words/New terms	• **Reproduction:** The biological process by which organisms produce offspring, ensuring the continuation of their species. • **Reproductive system:** This is the biological system responsible for producing gametes, facilitating fertilization, supporting pregnancy, and enabling the continuation of the species. • **Fertilization:** This is the process where a sperm cell from a male fuse with an egg cell from a female to form a zygote. • **Gametes:** Sex cell or reproductive cell containing only one set of dissimilar chromosomes, or half the genetic material necessary to form a complete organism. • **Pollination:** The process of transferring pollen from the male part of a flower to the female part, enabling fertilization and the production of seeds, fruits, and new plants.
Application	• **Agriculture:** Grafting & cutting (asexual) for faster crop propagation. • **Medicine**: Understanding human reproduction aids fertility treatments (IVF). • **Ecology**: Pollinators (bees, birds) maintain biodiversity by aiding plant reproduction. • **Conservation**: Artificial insemination helps endangered species breed. • **Food Production**: Hybrid crops (cross-pollination) improve yield and disease resistance.
Summary	• Reproduction is the process of producing offspring, occurring through sexual (fusion of male/female gametes) or asexual (no gametes) methods. • Humans have specialized reproductive systems (testes/ovaries), while flowering plants use pollination (self/cross) via wind, insects, or animals. • Asexual reproduction includes binary fission, budding, and vegetative propagation. • Both methods ensure species survival and genetic diversity.

What I have learnt?	
Skills Acquired	
My Question(s)/ Reflections	i. ii. iii. iv. v.
Action goal	

WEEKS TEN AND ELEVEN
REPRODUCTION
Reproduction is the ability of an organism to produce a new organism or offspring of their own kind.

Types of Reproduction
1. **Sexual reproduction:** is the coming together of male and female sex cells to form a new organism called zygote. It occurs in both plant and animals.
2. **Asexual reproduction:** is a type of reproduction which does not involve the coming together male and female sex cells. It occurs in plants and lower animals.
Examples:
a. Binary fission in amoeba, paramecium etc.
b. Vegetative propagation in high plants like cutting, budding, grafting, layering etc.
c. Spore formation in fungi
d. fragmentation in spirogyra
e. budding in hydra

Male Reproductive System in Man
The reproductive system consists of the following organs:

Reproductive Organ	Function
Testes (Testicles)	Produce sperm cells and the hormone testosterone.
Scrotum	A sac of skin that holds the testes outside the body and regulates their temperature.
Epididymis	Coiled tube where sperm mature and are stored.
Vas Deferens (Sperm Duct)	Transports sperm from the epididymis to the urethra.
Seminal Vesicles	Produce a fluid rich in nutrients that nourishes sperm.
Prostate Gland	Adds alkaline fluid to semen, helping sperm survive.
Bulbourethral (Cowper's) Glands	Secrete lubricating fluid that neutralizes acidity in the urethra.
Urethra	Passage through which semen and urine leave the body.
Penis	Organ that delivers semen into the female reproductive tract during reproduction.

The Female Reproductive System.
This consist of:

Reproductive Organ	Function
Ovaries	Produce eggs (ova) and hormones such as estrogen and progesterone.
Fallopian Tubes (Oviducts)	Transport the egg from the ovary to the uterus; fertilization usually occurs here.
Uterus (Womb)	Muscular organ where a fertilized egg implants and develops into a foetus.
Endometrium	Inner lining of the uterus that thickens each month and supports pregnancy.
Cervix	Lower narrow part of the uterus that connects to the vagina.

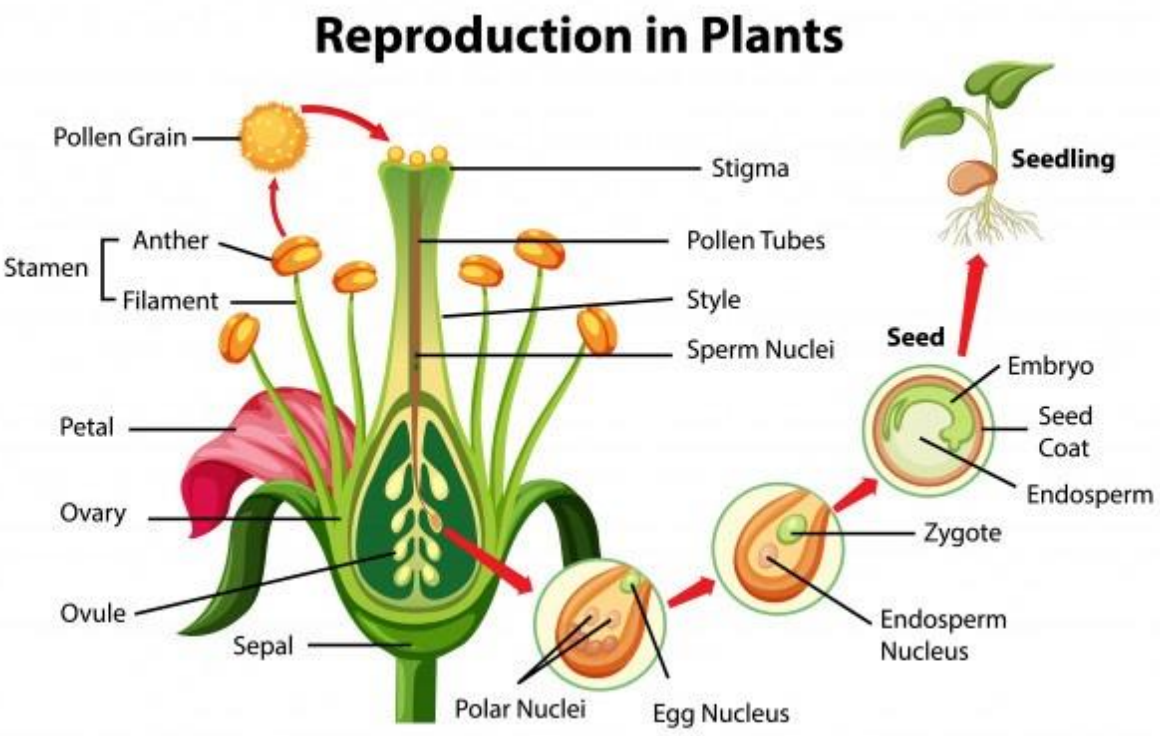
Vagina	Muscular canal that receives sperm, serves as the birth canal, and allows menstrual flow to exit the body.
Vulva	External female genital structures that protect the reproductive openings.

Sexual reproduction in flowering plants

The flowers constitute the reproduction organs in flowering plants.

MALE		
Organ	**Part**	**Function**
Stamen	Anther	Produces and stores pollen grains (male reproductive cells).
	Filament	Supports and holds the anther upright.
FEMALE		
Organ	**Part**	**Function**
Pistil/Carpel	Stigma	Receives pollen during pollination.
	Style	Connects the stigma to the ovary and provides a pathway for the pollen tube.
	Ovary	Contains ovules and develops into a fruit after fertilization.

The process by which plants reproduce sexually is called pollination. Pollination is the transfer of pollen grains from the anther to the stigma.



Types of pollination

1. **Self-pollination:** is the transfer of pollen grains from the anther of a flower to the stigma of the same flower or that of another flower of the same species.

2. **Cross pollination:** is the transfer of pollen grains from the anther of a flower to the stigma of a flower of another plant of the same species

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Agents of pollination

1. Wind
2. Water
3. Insects
4. Other animals